

Question Paper

Exam Date & Time: 20-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Semantic Web [CSE_4007]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Analyze and examine how ontology and metadata function together within the Semantic Web and inspect the relationships between ontology elements such as classes, properties, and concepts. (5)
A) Course Outcome: "CO1" / Cognitive Level: "Analysis"
- B) Organize the features of Web 1.0, Web 2.0, and Web 3.0 into a structured comparison that highlights their goals and technologies. Develop a conceptual model to show how decentralization influences the structure and information flow of the Semantic Web. (5)
Course Outcome: "CO2" / Cognitive Level: "Application"
- 2) Organize the categories of ontologies according to Guarino's classification (4)
A) Course Outcome: "CO2" / Cognitive Level: "Application"
- B) Summarize the characteristics of lightweight and heavyweight ontologies. (2)
Course Outcome: "CO2" / Cognitive Level: "Understand"
- C) Analyze the requirements of ontology languages. (4)
Course Outcome: "CO2" / Cognitive Level: "Analysis"
- 3) Apply the concept of medical ontology and explain their aim, the problems they solve, and the process they ensure. (3)
A) Course Outcome: "CO3" / Cognitive Level: "Application"
- B) Analyze the two processes involved in the Uschold and King's method. (5)
Course Outcome: "CO4" / Cognitive Level: "Analysis"
- C) Draw the block diagram showing the main processes of the Uschold and King's method. (2)
Course Outcome: "CO4" / Cognitive Level: "Understand"
- 4) Analyze and differentiate the fundamental modeling goals and structural approaches of the two specialized engineering ontologies, EngMath and PhysSys. (5)
A) Course Outcome: "CO3" / Cognitive Level: "Analysis"

- B) Consider a Multi-Agent System (MAS) designed for distributed problem solving, involving two primary agents operating in an e-commerce platform: (5)

1. Search Agent (A1): Responsible for fetching and interpreting product availability and price from external sources.
2. Negotiation Agent (A2): Responsible for setting transaction terms and acting on behalf of the user in a multi-agent transaction

Apply the principles of Agent Communication Language, Ontology based Agent Interaction and the architecture of Multi-Agent system to specify necessary five steps where these two agents communicate and coordinate to fulfill a user's request to "Find and reserve the lowest-priced item."

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5) Analyze the dependency of the Friend of a Friend (FOAF) vocabulary on the underlying structure of RDF (Resource Description Framework) to achieve its primary purpose using a suitable example. (5)

A)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- B) Analyze how semantic web technologies (such as RDF and OWL) enhance the reasoning and decision-making capabilities of agents in an agent-based semantic system. Support your answer with a suitable example scenario. (5)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

-----End-----

Event Name	Subject
------------	---------

058-END SEMSTER-ODD-2025 CSE_4007:Semantic Web

#	Course Outcome	Description	Marks
1	1	CO 1	5
2	2	CO 2	15
3	3	CO 3	13
4	4	CO 4	17
Total			50

* **NA** - CO Not Selected

Event Name	Subject
------------	---------

058-END SEMSTER-ODD-2025 CSE_4007:Semantic Web

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	2	4(8%)
3	Application	4	17(34%)
4	Analysis	6	29(58%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		12	50 (100%)